NIKIL NANDHA SELVARAJ

nikilselvaraj003@gmail.com | +1 (469) 487-1321 | Seattle, WA | LinkedIn | GitHub | Portfolio

SKILLS

Systems & Infrastructure: Linux Systems Engineering, Yocto-Based Embedded Linux & Firmware Development, Distributed Systems, Secure Boot & Firmware Signing • Cloud: AWS (Lambda, Step Functions, S3, CodePipeline, CodeCommit, Redshift, QuickSight) • Web Technologies: Angular, React.js, React Native, Node.js, HTML, CSS, RESTful Web Services • Languages: Python, C, C++, Java, JavaScript, C#, .NET • Databases: SQL Server, MySQL, MongoDB

WORK EXPERIENCE

Software Development Engineer (L5) | Amazon | Seattle, WA

June 2024 – Present

- Led the migration from legacy ONIE to a next-generation Linux-based bootstrap OS across a global fleet of 2.5 million network switches; designed and executed the rollout strategy that achieved 99.65% success rate with zero production incidents, enabling 100,000+ nodes to upgrade safely and transparently.
- Architected a CI/CD pipeline that automated weekly bootstrap OS releases; shifted integration testing left to give hardware vendors immediate failure feedback instead of discovering issues at deploy time, reducing release cycle time and enabling safe fleet-wide bootloader upgrades.
- Architected a Secure Boot system for firmware image signing and verification, integrating a centralized signing service with the Yocto build pipeline; ensured only authorized firmware can execute across production environments by cryptographically signing all firmware artifacts—preventing unauthorized code from running on deployed devices.
- Designed and shipped end-to-end observability for the device provisioning pipeline, enabling the org to track and diagnose fleet-wide rollout health in real time; the dashboard is now the standard source for fleet lifecycle metrics, replacing ad-hoc manual tracking and enabling faster incident response.
- Supported on-call engineering rotations and actively debugged complex system-level anomalies, resolving device-level failures and triaging provisioning issues; established debugging patterns and playbooks that reduced mean time to recovery for fleet-wide deployment issues.
- Mentored an intern through designing and shipping a unified release metrics tracking system that consolidated provisioning and rollout data across multiple storage systems; the system tracks fleet-wide deployment health and is now adopted by the release engineering team as the standard for monitoring rollout success metrics.

Software Development Engineer Intern | Amazon | Seattle, WA

Aug 2022 – Dec 2022

- Pre-Boot Error Logging: Engineered a hardware error logging mechanism within the open-source U-Boot bootloader; successfully captured early-stage initialization failures and passed critical telemetry up to the primary proprietary network OS for enhanced triage and remediation.

Associate Software Engineer | DUCEN | Chennai, India

Feb 2019 – July 2021

- Full-Stack Development: Engineered core features for a Business Intelligence platform used by internal analytics teams, collaborating across frontend and backend layers using Angular, C#, .NET, and MongoDB to deliver real-time reporting capabilities.
- Performance Optimization: Optimized data visualization by implementing lazy loading via d3.js, reducing page rendering latency by 28%.
- Payload Optimization: Achieved a 40% reduction in application payload size utilizing Brotli .NET compression for faster build file transfers.

EDUCATION

Master of Science: Computer Science | University of Texas at Arlington | GPA: 3.9/4.0 | Dec 2023

Bachelor of Engineering: Computer Science and Engineering | Anna University | Chennai, India | May 2019

PROJECTS

- **Real-time File Sharing: System Synchronization:** Built a multithreaded file sharing platform using React.js and Java, utilizing the file.WatchService API to keep on-device files in real-time sync with the cloud.
- **SecureGroup Chat App: Secure Communication:** Designed and developed a cross-platform messaging application featuring end-to-end encryption for secure file sharing and communication using React Native and Firebase.